

Literature Benchmark Brief: Liu_2024_ICLR

Title: iTransformer: Inverted transformers are effective for time series forecasting

Authors: Y. Liu, T. Hu, H. Zhang, H. Wu, S. Wang, L. Ma, M. Long (2024)

Venue: ICLR | **DOI:** 10.48550/arXiv.2310.06625

Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Inverted attention across variate tokens rather than temporal tokens

Empirical Scale & Experimental Dataset

Standard multivariate time series benchmarks

Relevance to IEEE TSTE & Sustainable Energy Systems

Top competitive deep learning baseline on WTB (RMSE 224.34)

Identified Gaps & Operational Vulnerabilities

Global attention fails to enforce local physical aerodynamic transitions

Role in Manuscript Defense & Reviewer #2 Proof

Headline baseline against which STGQ's parameter efficiency and tail compliance are audited